

AquaMind AI

Technical & Prototype Brief

Predictive water-network intelligence for earlier anomaly visibility and better utility decisions.

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VERSION 1.0	DATE October 2026	CURRENT DATA Synthetic	NEXT PROOF Real utility pilot

Executive Summary

AquaMind AI is a water-network intelligence concept and working browser prototype designed to turn pressure and flow signals into earlier operational insight. The current demonstration shows the complete operator workflow - sensing-style inputs, anomaly scenarios, zone-level risk prioritization, trend visualization, and recommended action - using synthetic hydraulic data.

The project is intentionally positioned as pre-pilot. It does not claim field accuracy, production leak-localization performance, or verified water-loss reduction. The next technical milestone is a bounded utility pilot using historical or live network data to calibrate analytics, measure false alarms and missed events, and validate operator usefulness.

CURRENT STAGE Prototype	DATA STATUS Synthetic	DEPLOYMENT Not field-validated	NEXT MILESTONE Utility pilot
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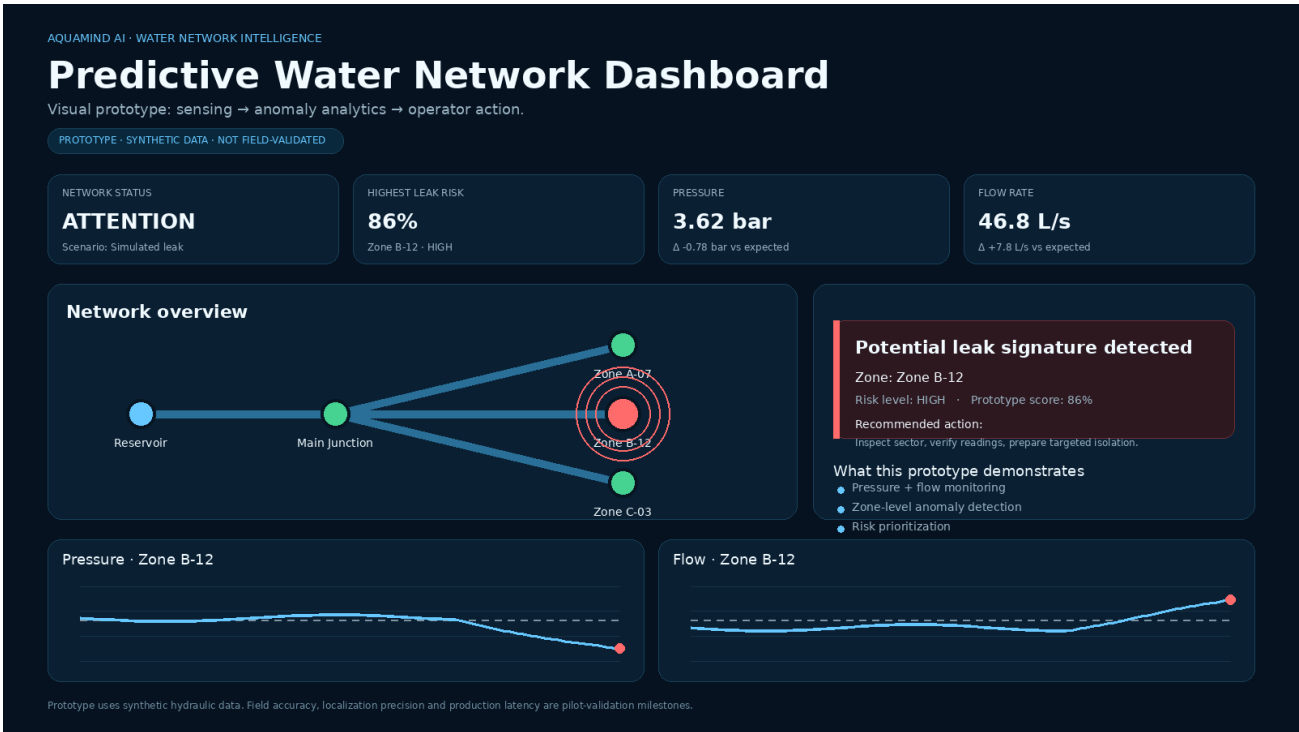


Figure 1. Current browser dashboard prototype. The displayed signals and events are synthetic and are used only to demonstrate the intended workflow.

1. System Architecture

The design separates what is implemented in the demonstration today from what must be validated in a real utility environment.

AquaMind AI - Technical Architecture

Clear separation between the current demonstration layer and the target field-pilot architecture.

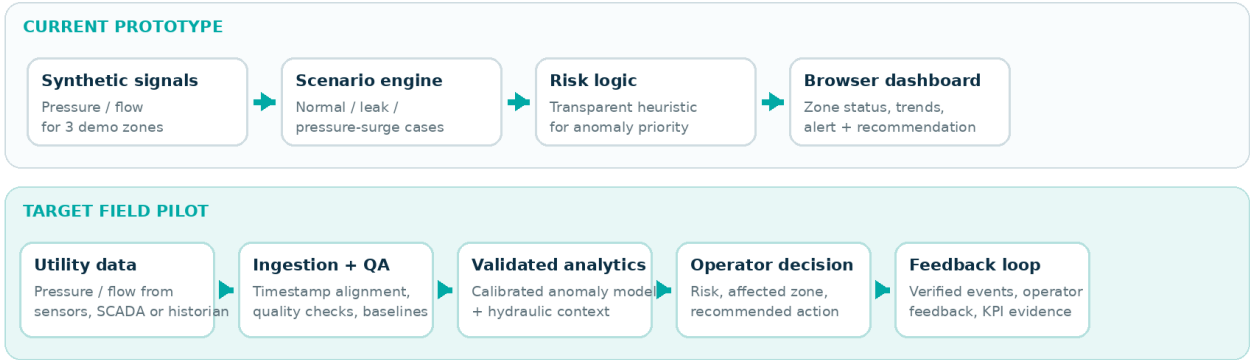


Figure 2. Current demonstrator flow versus target field-pilot architecture.

2. Current Prototype Scope

Capability	Current state	Technical note
Browser dashboard	Implemented	Streamlit interface with zone status, network view, trends, alerts, and operator recommendations.
Synthetic network scenarios	Implemented	Normal operation, simulated leak, and pressure-surge scenarios across three demo zones.
Risk prioritization	Implemented for demo	Transparent rule-based scoring combines pressure-drop and flow-rise deviations for prioritization.
Real sensor / SCADA integration	Not yet validated	Planned for pilot through a bounded read-only data connection or exported historian data.
Production AI model	Not field-calibrated	Future analytics require training/calibration and validation on verified utility events.
Autonomous valve control	Not in current prototype	Any control integration should follow technical validation, utility approval, and operational safeguards.

3. Prototype Technology Stack

Component	Current implementation
Software stack	Python, Streamlit, Pandas, NumPy, and Plotly
Demo engine	Programmatically generated synthetic pressure/flow series plus a transparent rule-based anomaly-priority heuristic for demonstration only

4. Pilot Validation Plan

The purpose of the first field pilot is to replace simulated evidence with measured technical and operational evidence. Success criteria should be agreed with the utility before the pilot starts.

Phase	Primary objective	Evidence to collect	Exit gate
1. Data access & audit	Connect historical or live pressure/flow data from a bounded zone.	Sensor coverage, sampling rate, completeness, missing data, known incidents.	Data quality is sufficient for repeatable analysis.
2. Analytics validation	Compare risk signals against verified events and normal operating periods.	Precision, recall, false positives, missed events, lead time.	Performance is reproducible and utility-relevant.
3. Operator workflow	Test whether alerts improve inspection and response decisions.	Alert usefulness, response time, operator feedback, action quality.	Operators can interpret and act on the output.
4. Deployment readiness	Package the validated workflow into a repeatable deployment.	Integration effort, uptime, support load, cost, commercial feedback.	Clear go/no-go case for scale-up.

5. Core Validation KPIs

KPI	Why it matters
Detection precision / positive predictive value	Shows how many alerts correspond to verified abnormal events.
Recall / missed-event rate	Measures whether the system detects events that matter to the utility.
False-positive rate	Determines alarm burden and operator trust.
Lead time	Measures how early a useful signal appears before visible failure or escalation.
Zone / DMA localization error	Measures whether the system points operators to the correct operational area.
Operator response time and action usefulness	Measures whether the interface improves real decisions, not only model scores.
Deployment friction	Tracks integration time, data readiness, support effort, uptime, and cloud/compute cost.

6. Recommended Pilot Safeguards

- **Read-only first:** Begin with historian or mirrored sensor feeds before any control-system integration.
- **Human-in-the-loop:** Treat AquaMind alerts as decision support during validation; operators retain authority.
- **Traceability:** Log data quality, model/risk outputs, operator decisions, and verified event outcomes.
- **Pre-agreed thresholds:** Define alarm thresholds, KPI targets, and escalation procedures jointly with the pilot utility.

7. Development Roadmap

Horizon	Technical focus	Primary deliverable	Evidence produced
0-3 months	Pilot design + data readiness	Bounded test zone, data dictionary, baseline analysis, integration plan.	Validated data pipeline and agreed evaluation protocol.
4-8 months	Field trial + model calibration	Calibrated analytics running against utility data with operator feedback.	Measured accuracy, false-alarm rate, lead time, and usability results.
9-12 months	Repeatable deployment package	Multi-zone deployment playbook, monitoring, support, and commercial package.	Evidence for scale-up and first recurring software deployment.

8. Deployment Requirements

- **Data access:** Timestamped pressure and flow data, zone/DMA identifiers, sensor metadata, and verified incident records where available.
- **Operational context:** Expected ranges, pump/valve events, maintenance windows, demand patterns, and known abnormal operating conditions.
- **Integration:** Initial read-only connection or secure data export; production integration requirements to be defined after pilot evidence.
- **Stakeholders:** Utility operations contact, hydraulic/domain advisor, data/integration contact, and field-validation partner.

9. Technical Risks & Current Limitations

Risk / limitation	Why it matters	Mitigation path
Synthetic-data bias	Demo scenarios may be cleaner than real hydraulic behavior.	Calibrate and test on real normal periods and verified incidents.
Sensor quality / gaps	Missing, noisy, or poorly synchronized data can create false alerts.	Data-quality checks, sensor-health rules, and confidence scoring.
Operational variability	Demand, pumps, valves, and maintenance can resemble anomalies.	Use operational context and hydraulic/domain constraints.
Model trust	Opaque alerts can be ignored even when technically accurate.	Provide explainable risk factors, affected zone, trend context, and operator feedback.
Control safety	Automated actuation can introduce operational risk.	Keep early pilots human-in-the-loop; add control only after validation and approvals.

10. Project Ownership & Positioning

AquaMind AI is currently founder-led by Mohammed Tariq Al-Saqqaf, a Mechatronics Engineer developing the prototype, technical architecture, and pilot-validation pathway. The project is seeking utility access, hydraulic-domain mentorship, field-integration support, and commercial validation partners.

CURRENT POSITIONING

Functional prototype and pilot-ready concept - not yet a field-validated commercial product. Field accuracy, leak-localization performance, water-loss reduction, and utility ROI must be demonstrated through real-world validation before being presented as proven outcomes.

Document basis: AquaMind AI prototype implementation, prototype README/demo notes, and founder-provided DMZ pitch materials.